

Assume the following throughout this entire sheet

- $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ a vector-valued function in $\mathbf{R}^3$ with continuous partial derivatives.
- $\mathbf{G}(x, y) = [M(x, y), N(x, y)]$ a vector-valued function in $\mathbf{R}^2$ with continuous partial derivatives.
- $g(x, y, z)$ a scalar-valued function with continuous first partial derivatives
- $\mathbf{r}(t)$ represents some parameterization of a curve C for $a \leq t \leq b$
- C_0 is a simple closed curve enclosing a region R

Operators:

- $\nabla g(x, y, z) = [g_x, g_y, g_z]$
- $\operatorname{div} \mathbf{F} = M_x + N_y + P_z$
- $\operatorname{curl} \mathbf{F} = [P_y - N_z, M_z - P_x, N_x - M_y]$

Line Integrals:

- Vector-Valued: $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$
- Scalar-Valued: $\int_C g ds = \int_a^b g(\mathbf{r}) \cdot |\mathbf{r}'(t)| dt$

Green's Theorem: $\oint_{C_0} \mathbf{G} \cdot d\mathbf{r} = \iint_R (\operatorname{curl} \mathbf{G}) \cdot \mathbf{k} dA$